

practice, one should consider this “robustness” versus “efficiency” trade-off when choosing between θ_{es}^{bal} and θ_{es} .

Remark 8. We note that $\theta_{es}^{bal}(e; e')$ closely resembles the empirical practice of only reporting event-study-type coefficients for the event periods that do not suffer from compositional changes, see, e.g., [McCrory \(2007\)](#) and [Bailey and Goodman-Bacon \(2015\)](#). An important caveat is that our proposed event-study-type estimands $\theta_{es}^{bal}(e; e')$ are not based on dynamic TWFE specifications akin to (3.3), and therefore bypass the pitfalls associated with (3.3) highlighted by [Sun and Abraham \(2020\)](#).

3.1.2 How do average treatment effects vary across groups?

It is also straightforward to aggregate our group-time average treatment effects to understand heterogeneity in the effect of participating in the treatment across groups. Although understanding this sort of heterogeneity is relatively less common in applied work than trying to understand dynamic effects as discussed above, there are still a number of cases in economics where understanding this sort of heterogeneity may be of interest. For example, work on the effect of graduating during a recession on labor market outcomes ([Oreopoulos et al. \(2012\)](#)) or the effect of job displacement across the business cycle ([Farber \(2017\)](#)) are related to heterogeneous effects across groups. More generally, these parameters are useful for understanding if the effect of participating in the treatment was larger for groups that are treated earlier relative to groups that are treated later. In addition, in the next section, these parameters will be the building block for our main measure of the overall effect of participating in the treatment. To consider heterogeneous effects across groups, we consider the following parameter

$$\theta_{sel}(\tilde{g}) = \frac{1}{\mathcal{T} - \tilde{g} + 1} \sum_{t=\tilde{g}}^{\mathcal{T}} ATT(\tilde{g}, t). \quad (3.7)$$

$\theta_{sel}(\tilde{g})$ is the average effect of participating in the treatment among units in group $\tilde{g}$, across all their post-treatment periods.

3.1.3 What is the cumulative average treatment effect of the policy across all groups until time $\tilde{t}$?

In some applications, researchers may want to construct an aggregated target parameter to highlight treatment effect heterogeneity with respect to calendar time. In economics, for example, researchers might wish to study heterogeneous treatment effects across the business cycle. The average effect of participating in the treatment in time period t (across groups that have adopted the treatment by period t) is given by

$$\theta_c(\tilde{t}) = \sum_{g \in \mathcal{G}} \mathbf{1}\{\tilde{t} \geq g\} P(G = g | G \leq \tilde{t}) ATT(g, t) \quad (3.8)$$

An extension to this parameter is to think about the *cumulative* effect of participating in the treatment up to some particular time period. For instance, in active labor market applications, policy makers may want to know the cumulative average effect of a given training program on earnings from the year that the first group of people were trained until year $\tilde{t}$. This would provide a measure of the cumulative

earnings gains induced by the training program. Alternatively, in health applications, researchers may want to measure how many COVID-19 cases have been averted by shelter-in-place orders up to day $\tilde{t}$. To consider the cumulative effect, consider the following parameter

$$\theta_c^{cumu}(\tilde{t}) = \sum_{t=2}^{\tilde{t}} \theta_c(t). \quad (3.9)$$

$\theta_c^{cumu}(\tilde{t})$ can be interpreted as the cumulative average treatment effect among the units that have been treated by time $\tilde{t}$.

3.2 Aggregations into Overall Treatment Effect Parameters

Finally in this section, we consider some ideas for aggregating group time average treatment effects into an overall effect of participating in the treatment. One very simple idea is to just average all of the identified group-time average treatment effects together; i.e., to consider the parameter

$$\theta_W^O = \frac{1}{\kappa} \sum_{g \in \mathcal{G}} \sum_{t=2}^{\mathcal{T}} \mathbf{1}\{t \geq g\} ATT(g, t) P(G = g | G \leq \mathcal{T}) \quad (3.10)$$

where $\kappa = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\mathcal{T}} \mathbf{1}\{t \geq g\} P(G = g | G \leq \mathcal{T})$ (which ensures that the weights on $ATT(g, t)$ in the second term sum up to one). θ_W^O is a weighted average of each $ATT(g, t)$ putting more weight on $ATT(g, t)$'s with larger group sizes. Unlike β in the TWFE regression specification (3.2), this simple combination of $ATT(g, t)$'s immediately rules out troubling issues due to negative weights; as a particular example, when the effect of participating in the treatment is positive for all units, this aggregated parameter cannot be negative.

That being said, just requiring positive weights is a very minimal requirement of a reasonable overall treatment effect parameter. For example, one drawback of θ_W^O is that it systematically puts more weight on groups that participate in the treatment for longer. Instead, we suggest the following parameter as a general-purpose summary of the average effect of participating in the treatment

$$\theta_{sel}^O = \sum_{g \in \mathcal{G}} \theta_{sel}(g) P(G = g | G \leq \mathcal{T}) \quad (3.11)$$

where $\theta_{sel}(g)$ is the average effect of participating in the treatment for units in group g as defined in Equation (3.7) above. θ_{sel}^O first computes the average effect for each group (across all time periods) and then averages these effects together across groups to summarize the overall average effect of participating in the treatment. Thus, θ_{sel}^O is the average effect of participating in the treatment experienced by all units that ever participated in the treatment. In this respect, its interpretation is the same as the ATT in the canonical DiD setup with two periods and two groups. This is an attractive property for a summary measure of the overall effect of participating in the treatment in the context of multiple time periods and variation in treatment timing.

Working by analogy, one can also define overall treatment effect parameters by averaging $\theta_{es}(e)$ across

all event times or $\theta_c(t)$ across all time periods, i.e.,

$$\theta_{es}^O = \frac{1}{\mathcal{T}-1} \sum_{e=0}^{\mathcal{T}-2} \theta_{es}(e) \quad \theta_c^O = \frac{1}{\mathcal{T}-1} \sum_{t=2}^{\mathcal{T}} \theta_c(t) \quad (3.12)$$

In our view, the appeal of these aggregations is likely to be somewhat more limited than that of θ_{sel}^O in most applications. For example, the interpretation of θ_{es}^O is complicated by the issue of the changing composition of groups across different values of e discussed above (similar arguments apply to θ_c^O as well). As before, one can circumvent the issue of the changing composition of groups by balancing the sample with respect to event time. A (local) single summary parameter is given by

$$\theta_{es}^{O,bal}(e') = \frac{1}{e'+1} \sum_{e=0}^{e'} \theta_{es}^{bal}(e, e') \quad (3.13)$$

This is the average effect of participating in the treatment over the first e' periods of exposure to the treatment. This is also a reasonable alternative overall treatment effect parameter, but it should also be noted that it is local to groups that participated in the treatment for at least e' periods.

As a final comment, in general, none of the overall effect parameters considered in this section are equal to each other except in the special case where $ATT(g, t)$ is the same for all groups and all time periods. In that case, all of the aggregated parameters, including β from the TWFE regression, are equal to each other.

4 Estimation and Inference

So far we have focused on the identification and aggregation stages of the analysis. In this section, we show how one can build on these results to form estimators for and conduct inference about the group-time average treatment effects and their summary measures described in Section 3. Given that the $ATT(g, t)$'s are the main building blocks of our analysis, we start with them.

First, it is important to notice that our identification results in Theorem 1 are constructive and suggest a simple and intuitive two-step estimation strategy to estimate the $ATT(g, t)$'s. In the first step, one estimates the nuisance functions for each group g and time period t — $p_g(x)$ and/or $m_{g,t,\delta}^{nev}(X)$ if one relies on Assumption 4, and $p_{g,t+\delta}(x)$ and/or $m_{g,t,\delta}^{ny}(X)$ if one relies on Assumption 5. In the second step, one plugs the fitted values of these estimated nuisance functions into the sample analogue of the considered $ATT(g, t)$ estimand to obtain estimates of the group-time average treatment effect.

A natural question that then arises is which type of approach one should use in practice: the outcome regression, inverse probability weighting, or the doubly-robust one. Although these three different approaches are equivalent from the *identification* perspective, this is not the case from the *estimation/inference* perspective. The OR approach requires researchers to *correctly* model the outcome evolution of the comparison group to estimate the group-time average treatment effects. This approach is explicitly connected with the conditional parallel trends assumption required in DiD analysis as this condition is usually expressed in terms of conditional expectations. The IPW approach, on the other hand, avoids explicitly modeling the outcome evolution of the comparison group and therefore does not

rely on putative model restrictions directly tied to the parameter of interest. Instead, the IPW approach requires one to *correctly* model the conditional probability of unit i being in group g given their covariates X and that they are either in group g or in an appropriate comparison group. The DR approach combines both the OR and IPW approaches as it relies on modeling both the outcome evolution and the propensity score. However, it only requires one to correctly specify *either (but not necessarily both)* the outcome evolution for the comparison group or the propensity score model (Sant’Anna and Zhao, 2020). Thus, the DR approach enjoys additional robustness against model misspecifications when compared to the OR and IPW approaches. In addition, the DR approach potentially allows one to use a broader set of estimation methods such as those that involve penalization and some types of model selection, see, e.g. Belloni et al. (2017).

Given these attractive robustness features associated with the DR approach, in this section we consider estimators of the DR form; the discussion on how to proceed with the OR and IPW approaches is analogous and therefore omitted. We also focus on parametric estimators for the nuisance functions. We consider this case mainly for its practical appeal which is especially true in applications where the number of covariates is fairly large and the number of observations is only moderate.¹³

More concisely, let

$$\widehat{ATT}_{dr}^{nev}(g, t; \delta) = \mathbb{E}_n \left[(\widehat{w}_g^{treat} - \widehat{w}_g^{comp, nev}) \left(Y_t - Y_{g-\delta-1} - \widehat{m}_{g,t,\delta}^{nev} \left(X; \widehat{\beta}_{g,t,\delta}^{nev} \right) \right) \right], \quad (4.1)$$

$$\widehat{ATT}_{dr}^{ny}(g, t; \delta) = \mathbb{E}_n \left[(\widehat{w}_g^{treat} - \widehat{w}_g^{comp, ny}) \left(Y_t - Y_{g-\delta-1} - \widehat{m}_{g,t,\delta}^{ny} \left(X; \widehat{\beta}_{g,t,\delta}^{ny} \right) \right) \right], \quad (4.2)$$

where

$$\widehat{w}_g^{treat} = \frac{G_g}{\mathbb{E}_n[G_g]}, \quad \widehat{w}_g^{comp, nev} = \frac{\frac{\widehat{p}_g(X; \widehat{\pi}_g) C}{1 - \widehat{p}_g(X; \widehat{\pi}_g)}}{\mathbb{E}_n \left[\frac{\widehat{p}_g(X; \widehat{\pi}_g) C}{1 - \widehat{p}_g(X; \widehat{\pi}_g)} \right]}, \quad \widehat{w}_g^{comp, ny} = \frac{\frac{\widehat{p}_{g,t+\delta}(X; \widehat{\pi}_{g,t+\delta}) (1 - D_{t+\delta}) (1 - G_g)}{1 - \widehat{p}_{g,t+\delta}(X; \widehat{\pi}_{g,t+\delta})}}{\mathbb{E}_n \left[\frac{\widehat{p}_{g,t+\delta}(X; \widehat{\pi}_{g,t+\delta}) (1 - D_{t+\delta}) (1 - G_g)}{1 - \widehat{p}_{g,t+\delta}(X; \widehat{\pi}_{g,t+\delta})} \right]},$$

with $\widehat{p}_g(\cdot; \widehat{\pi}_g)$, $\widehat{p}_{g,t+\delta}(\cdot; \widehat{\pi}_{g,t+\delta})$, $\widehat{m}_{g,t,\delta}^{nev}(\cdot; \widehat{\beta}_{g,t,\delta}^{nev})$ and $\widehat{m}_{g,t,\delta}^{ny}(\cdot; \widehat{\beta}_{g,t,\delta}^{ny})$ being (parametric) estimators of $p_g(\cdot)$, $p_{g,t+\delta}(\cdot)$, $m_{g,t,\delta}^{nev}(\cdot)$ and $m_{g,t,\delta}^{ny}(\cdot)$, respectively, and for a generic Z , $\mathbb{E}_n[Z] = n^{-1} \sum_{i=1}^n Z_i$. $\widehat{ATT}_{dr}^{nev}(g, t; \delta)$ and $\widehat{ATT}_{dr}^{ny}(g, t; \delta)$ are our proposed DR DiD estimators for $ATT(g, t)$ when one invokes Assumption 4 and Assumption 5, respectively. These estimators extend the DR DiD estimators of Sant’Anna and Zhao (2020) from the two periods, two groups setup to the multiple groups, multiple periods setup while allowing for possible treatment anticipation. In addition, these estimators are of the Hájek (1971)-type and their associated weights are guaranteed to sum up to one in finite samples. As illustrated by Busso et al. (2014), this usually leads to improved finite sample properties.

With the estimators for the $ATT(g, t)$ ’s in hand, one can use the analogy principle and combine these

¹³Alternatively, one could adopt a fully nonparametric approach. Let $f(x)$ be a generic notation for the nuisance functions. From Newey (1994), Chen et al. (2003), Ai and Chen (2003, 2007, 2012), and Chen et al. (2008), one can see that the use of nonparametric first-step estimators $\widehat{g}(x)$ of $g(x)$ is warranted provided that $\|\widehat{g}(x) - g(x)\|_{\mathcal{H}} = o_p(n^{-1/4})$ for a pseudo-metric $\|\cdot\|_{\mathcal{H}}$, $\mathcal{H}$ being a vector space of functions. However, when the dimension of X is moderate or large, as is often the case in empirical applications, conditions ensuring that $\|\widehat{g}(x) - g(x)\|_{\mathcal{H}} = o_p(n^{-1/4})$ can be rather stringent due to the so-called “curse of dimensionality”.